CHIMA CYRIL HAMPO

Building-Energy & Heat-Health Computational Scientist — who ships production software

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SUMMARY

PhD computational scientist (Architectural Engineering, Drexel, 2026; NIH-funded) working at the intersection of **building-energy simulation, indoor heat/humidity exposure, and climate-health equity** — and one of the few in this space who also **builds and ships full-stack software**. Author of **20+ peer-reviewed papers** (multiple first-author) with work **featured in The New York Times**. I turn research-grade methods (EnergyPlus/ResStock-style stock modeling, uncertainty/sensitivity analysis, Python/R pipelines) into production-minded decision tools — most recently **ReplocX**, a deployed React + FastAPI decision-support application.

CORE SKILLS

Building & energy simulation: EnergyPlus, OpenStudio, DesignBuilder, Insight, Revit; overheating/thermal-comfort assessment; indoor temperature & humidity prediction; scenario analysis across typologies, climate zones, code vintages.
Programming & data science: Python, R, SQL, MATLAB; data pipelines/ETL; machine-learning & forecasting workflows; uncertainty & sensitivity analysis; reproducible research.
Software engineering: React, TypeScript, Vite; FastAPI / Python services; REST + OpenAPI, SQLite; Docker, CI/CD, Playwright e2e; geospatial (ZIP/ZCTA/county/CBSA).
Sustainability analytics: Life-cycle assessment & costing (openLCA, Umberto LCA+, ecoinvent); techno-economic/scenario evaluation.

SELECTED PROJECTS

ReplocX — full-stack representative-location decision-support app (Python, FastAPI, React/TypeScript, Docker, CI/CD)

- Built an end-to-end web app selecting representative U.S. locations for national building-stock, climate, and heat-health simulation — deterministic scoring, allocation, ZIP explanation, weather-QC, scenario comparison, and exports.
- Engineered a reproducible pipeline over a ~4,019-candidate national universe resolving to 20 climate-region × urbanicity strata, behind a versioned **/api/v1** (OpenAPI, health probes, structured logs).
- Shipped with release maturity: unit + Playwright e2e tests, type/lint/build gates, Docker smoke tests, CSP/security headers, public-demo / private-data separation. Live demo deployed on Render.

Overheating in U.S. Residential Buildings — EnergyPlus/OpenStudio simulation quantifying indoor overheating across U.S. home types and climates for heat-health risk assessment (SimBuild 2026, IBPSA-USA).

AquaFlow — whole-building water management platform (HackSimBuild / IBPSA-USA · 24-hour hackathon · 2 awards)

- 🏆 Won **Most Popular Hack** and **Most Creative Demonstration**. Led the AI & front-end layer — built an interactive React 3D scene (AI-accelerated) and wired live modeling calculations into the live demo.
- Real-time tradeoff comparison of rainwater harvesting, graywater recycling, and green roofs across water use, cost, and resilience.

EXPERIENCE

Graduate Research Assistant — Building Performance & Sustainability 2022–Present

Dept. of Civil, Architectural & Environmental Engineering, Drexel University (NIH Grant 1R01HL164726-01)

- Lead contributor developing scalable modeling to estimate indoor thermal/moisture conditions in U.S. residential buildings under varying climate and operational conditions.
- Developed, parameterized, and QA'd EnergyPlus/OpenStudio/DesignBuilder workflows comparing envelope and operational strategies across climate zones and building eras.
- Built reproducible Python/R pipelines to extract, clean, structure, and analyze large simulation datasets into publication-ready outputs.
- Applied uncertainty quantification and sensitivity analysis to characterize drivers of indoor temperature/humidity predictions.
- Collaborated across engineering and public health to translate building-science outputs into health-relevant exposure metrics; co-authored peer-reviewed reviews of indoor temperature and health.

Graduate Research Assistant — Sustainable Energy Systems 2019–2022

Dept. of Mechanical Engineering, Universiti Teknologi Petronas, Malaysia

- Performed thermodynamic modeling and system-performance analysis for district cooling / chiller configurations, including thermal energy storage integration.
- Conducted life-cycle assessment and life-cycle costing to evaluate environmental and economic trade-offs; published multiple peer-reviewed studies.

SELECTED PUBLICATIONS & RECOGNITION (20+ PEER-REVIEWED — FULL LIST ON GOOGLE SCHOLAR)

- Hampo, C.C., Schinasi, L.H., & Hoque, S. (2025). Indoor temperature & humidity prediction in residential buildings: white-box and black-box modeling. *Building and Environment*.
- Hampo, C.C., Schinasi, L.H., & Hoque, S. (2024). Surviving indoor heat stress in the United States. *Heliyon* 10(3).
- Edwards, J.R., De Roos, A.J., Hampo, C.C., et al. (2025). Residential indoor temperatures and health: a scoping review. *Science of the Total Environment* 979.
- **Research featured in The New York Times (2024)** on indoor overheating and climate resilience.

AWARDS · LEADERSHIP · EDUCATION

HackSimBuild (IBPSA-USA) — Most Popular Hack & Most Creative Demonstration (2026). PDC Student Design Challenge — 1st Runner-Up, national (2025). NIH Graduate Research Grant (2022–2027). Yayasan UTP Research Scholarship. **President, Drexel Graduate Student Association (2025–26)** (VP 2024–25). Member: ASHRAE, ASHE.

PhD, Architectural Engineering, Drexel University — Expected 2026 (Minor: Business Analytics) · **MSc, Mechanical Engineering (by Research)**, Universiti Teknologi Petronas — 2021 · **BEng, Mechanical Engineering**, University of Benin — 2017